

Cryptanalysis of Vigenere Ciphers

We consider **Ciphertext-Only** attacks.

Simplest model of the attacker, he only has Ciphertext

First we consider two techniques to compute the block size m .

Kasiski's Observation: Two identical pieces of plaintext will be encrypted to the same ciphertext whenever their occurrence of plaintext is x positions apart, where $x = 0 \bmod m$.

Kasiski's Test: Search the ciphertext for pairs of identical strings of length at least 3.

Record distances between the starting positions of the two segments.

If we record several such distances $d_1, d_2, \dots$ then we can conjecture that $m \mid \gcd(d_1, d_2, \dots)$.

Friedman's Test uses **Index of Coincidence**:

Let $I_c(x)$ = probability that two random elements of the n -letter string x are identical.

frequency of occurrence

Let $f_0, f_1, \dots, f_{25}$ be the # of occurrence of A, B, ..., Z, respectively in the ciphertext.

$$I_c(x) = \frac{\sum_{i=0}^{25} \binom{f_i}{2}}{\binom{n}{2}} = \frac{\sum_{i=0}^{25} f_i(f_i - 1)}{n(n - 1)} \quad (1)$$

$\frac{n!}{2!(n-2)!}$

Now recall the frequency table

Single	Frequency	
E	.127	13%
T	.091	9%
A	.082	
O	.075	
I	.070	
N	.067	
S	.063	
H	.061	6%

From this table we can get the expected value of $I_c(x)$; p_i^2 is the probability that in the string x both elements chosen at random are equal to the i -th letter.

Here, p_i be the expected probability of occurrence of the i -th letter in the English language.

Now English text and random strings differ:

Ciphertext	English Text	Random String
$I_c(x)$ in Formula (1)	$\sum_{i=0}^{25} p_i^2 = 0.065$	$26(1/26)^2 = 0.038$

take our ciphertext & break it into m columns

completely random string of letters

Friedman's Method: Guess m . Partition the ciphertext y into m columns $y_1, y_2, \dots, y_m$ each of length n/m .

If our guess m is the correct block size then $I_c(y_i) \approx 0.065$, while if m is not correct then $I_c(y_i) \approx 0.038$.

we will end up with Index of Coincidence similar

Example

Consider the ciphertext obtained from Vigenere cipher.

CHREEVOAHMAERATBIAXXWTNXBEEOPHBSBQMQEQRBW
RVXUOAKXAOSXXWEAHBWGJMMQMNKGRFVGXWTRZXWIAK
LXFPSKAUTEMNDCMGTSXMXBTUIADNGMGPSRELXNJELX
VRVPTULHDNQWTDYGBPHXTFALJHASVBFXNGLLCHR
ZBWELEKMSJIKNBHWRJGNMGJSGLXFEYPHAGNRBIEQJT
AMRVLCRREMNDGLXRRIMGNSNRWCHRQHAIEYEVTAQEBBI
PEEWEVKAKOEWADREMXTBHHCHRTKDNVRZCHRCLQOHP
WQAIIWXRNRMGWOIIFKEE

We will analyze this ciphertext. Our method will be the following.

1. We will make a guess for the block size and subsequently use, (a) Kasiski's method, and (b) Friedman's method to compute m . Both methods will give the same value for the block size.
2. Assuming now that m is known we will use frequency analysis and the mutual index of coincidence to decipher the text.

Kasiski's Method: The cipher text string CHR occurs in four places in ciphertext and at the positions:

1, 166, 236, 286.

The distances from the first occurrence are 165, 235, 285.

Since $\gcd(165, 235, 285) = 5$ our guess for the block size is 5.

key is of length 5

If we want to verify that this guess is correct, we use Friedman test.

Friedman's Method: We compute the index of coincidence for the given ciphertext.

For each guess m , we have blocks of size n/m . So we have block size 5 as the best guess.

<i>guess</i> m : key length	m	$I_c \leftarrow$ Sum of Probabilities
<u>1</u>	0.045	<i>(Closer to Random Value)</i>
<u>2</u>	0.046, 0.041	<i>doesn't help</i>
<u>3</u>	0.043, 0.050, 0.047	
<u>4</u>	0.042, 0.039, 0.046, 0.040	
<u>5</u>	0.063, 0.068, 0.069, 0.061, 0.072	

values close to 0.065

key is of length 5

Now we have fairly high confidence the key is of length 5

Assume we know the block size m

try to figure out the actual value of the key
Mutual index of coincidence: $MI_c(x, x')$ is

the probability that a random element of x is identical to a random element of x' .

Let $f_0, f_1, \dots, f_{25}, f'_0, f'_1, \dots, f'_{25}$ be the frequency of occurrence of $A, B, \dots, Z$, respectively in the strings x, x' . So we have the formula.

$$MI_c(x, x') = \frac{\sum_{i=0}^{25} f_i f'_i}{nn'} \quad (2)$$

Since we know m we can partition the ciphertext y into m blocks $y_1, \dots, y_m$.

Assuming $K = (k_1, k_2, \dots, k_m)$ is the key being used we can estimate $MI_c(y_i, y_j)$.

Take a random character in y_i and a random character in y_j . The probability that both characters are A is $p_{-k_i} p_{-k_j}$. The probability that both characters are B is $p_{1-k_i} p_{1-k_j}$, etc. Hence,

$$MI_c(\mathbf{y}_i, \mathbf{y}_j) \approx \sum_{h=0}^{25} p_{h-k_i} p_{h-k_j} = \sum_{h=0}^{25} p_h p_{h+k_i-k_j}.$$

Note that the mutual coincidence depends on the relative shift $k_i - k_j$. Moreover, the mutual coincidence of the relative shift $k_i - k_j$ is the same with the mutual coincidence of the relative shift $k_j - k_i$.

Relative Shift	Expected value of MI_c
0	0.065
1	0.039
2	0.032
3	0.034
4	0.044
5	0.033
6	0.036
7	0.039
8	0.034
9	0.034
10	0.038
11	0.045
12	0.039
13	0.043

Next we tabulate the relative shifts between 0 and 13.

The mutual coincidences vary from 0.031 to 0.045 if the relative shift is nonzero

The mutual coincidence is 0.065 if the relative shift is 0.

This observation is being used to formulate a likely guess for the shift $\ell = k_i - k_j$.

Now fix y_i . Let $y_j^0, y_j^1, y_j^2, \dots$ be the result of encrypting y_j by $e_0, e_1, e_2, \dots$

Tabulate the mutual indices $MI_c(y_i, y_i^g)$.

When $g = \ell$ the mutual index of coincidence should be 0.065.

When $g \neq \ell$ the mutual index of coincidence should be between 0.031 and 0.045.

Tabulate, by computer, the 260 values

$$MI_c(y_i, y_j^g), \quad 1 \leq i < j \leq 5, 0 \leq g \leq 25$$

For each (i, j) look at the values of $MI_c(y_i, y_j^g)$ which are close to 0.065. If the value is unique we conjecture it is the value of the relative shift.

Six such potential values are boxed in the table.

Pair	Relative Shift
(1, 2)	9
(1, 5)	16
(2, 3)	13
(2, 5)	7
(3, 5)	20
(4, 5)	11

The resulting equations are

$$\begin{aligned} k_1 - k_2 &= 9 & k_1 - k_5 &= 16 \\ k_2 - k_3 &= 13 & k_2 - k_5 &= 7 \\ k_3 - k_5 &= 20 & k_4 - k_5 &= 11 \end{aligned}$$

The key is $(k_1, k_1 + 17, k_1 + 4, k_1 + 21, k_1 + 10)$, for some $k_1 \leq 25$, which is a cyclic shift of *AREVK*. The key is easily found: *JANET*.

Observed Mutual Indices of Coincidence

i	j	value of $MI_c(y_i, y_j^g)$								
1	2	.028	.027	.028	.034	.039	.037	.026	.025	.052
		.068	.044	.026	.037	.043	.037	.043	.037	.028
		.041	.041	.034	.037	.051	.045	.042	.036	
1	3	.039	.033	.040	.034	.028	.053	.048	.033	.029
		.056	.050	.045	.039	.040	.036	.037	.032	.027
		.037	.036	.031	.037	.055	.029	.024	.037	
1	4	.034	.043	.025	.027	.038	.049	.040	.032	.029
		.034	.039	.044	.044	.034	.039	.045	.044	.037
		.055	.047	.032	.027	.039	.037	.039	.035	
1	5	.043	.033	.028	.046	.043	.044	.039	.031	.026
		.030	.036	.040	.041	.024	.019	.048	.070	.044
		.028	.038	.044	.043	.047	.033	.026	.046	
2	3	.046	.048	.041	.032	.036	.035	.036	.030	.024
		.039	.034	.029	.040	.067	.041	.033	.037	.045
		.033	.033	.027	.033	.045	.052	.042	.030	
2	4	.046	.034	.043	.044	.034	.031	.040	.045	.040
		.048	.044	.033	.024	.028	.042	.039	.026	.034
		.050	.035	.032	.040	.056	.043	.028	.028	
2	5	.033	.033	.036	.046	.026	.018	.043	.080	.050
		.029	.031	.045	.039	.037	.027	.026	.031	.039
		.040	.037	.041	.046	.045	.043	.035	.030	
3	4	.038	.036	.040	.033	.036	.060	.035	.041	.029
		.058	.035	.035	.034	.053	.030	.032	.035	.036
		.036	.028	.046	.032	.051	.032	.034	.030	
3	5	.035	.034	.034	.036	.030	.043	.043	.050	.025
		.041	.051	.050	.035	.032	.033	.033	.052	.031
		.027	.030	.072	.035	.034	.032	.043	.027	
4	5	.052	.038	.033	.038	.041	.043	.037	.048	.028
		.028	.036	.061	.033	.033	.032	.052	.034	.027
		.039	.043	.033	.027	.030	.039	.048	.035	

The almond tree was in tentative blossom. The days were longer often ending with magnificent evenings of corrugated pink skies. The hunting season was over ...